



The administration of *Sheng Hua Tang* immediately after delivery to reduce the incidence of retained placenta in Holstein dairy cows

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ABSTRACT

Sheng Hua Tang, a classical herbal formula consisting of *Radix Angelicae sinensis*, *Ligustici rhizoma*, *Semen persicae*, *Zingiberis rhizoma*, and *Radix glycyrrhizae*, is known to be beneficial in alleviating postpartum diseases and facilitating a return to normal reproductive function. This study investigated whether the administration of *Sheng Hua Tang* within 2 to 4 hours after delivery was effective as a preventive treatment for reducing the risk of retained placenta in Holstein dairy cows. A total of 357 cows, each of which had delivered its calf spontaneously, were randomly allocated to one of two groups. In the treatment group, the cows ($n = 175$) received *Sheng Hua Tang* with an oral dose of 0.36 g crude herb per kg·body weight once daily for three consecutive days. The controls ($n = 182$) received no treatment. The placental retention proportion was 4.0% and 17.0% within 12 hours after delivery in the treated and control animals, respectively ($P < 0.01$). We found decreases in the calving-to-first-service interval (73.2 ± 25.1 vs. 81.9 ± 32.8 days; $P < 0.01$), calving-to-conception interval (93.4 ± 38.8 vs. 114.6 ± 42.9 days; $P < 0.01$), and service per conception (1.5 ± 0.8 vs. 1.9 ± 1.0 days; $P < 0.01$) in the treatment group compared with the control group. The first artificial insemination conception proportion was higher in the treatment group than in the control group (60.4% vs. 41.1%; $P = 0.01$). Moreover, the between-group difference in the proportion of cows that were pregnant within 180 days postpartum approached statistical significance (88.2% vs. 80.6%; $P = 0.07$). *Sheng Hua Tang* showed beneficial effects in reducing the incidence of retained placenta and improving subsequent reproductive performance in cows. This preventive treatment strategy would be effective in improving the management of puerperal health. The potential benefits of *Sheng Hua Tang* warrant further investigation to determine whether this preventive treatment strategy can be endorsed as a general preventive approach in postpartum cows.

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1. Introduction

In dairy cattle, retained placenta can be considered a puerperal syndrome having a potentially multifactorial

etiology and posing a substantial risk factor for postpartum uterine diseases such as metritis and endometritis [1,2]. Consequently, a retained placenta is often associated with delayed uterine involution and subsequent impaired fertility [3–6].

Many common therapies have not been shown to be effective in facilitating the release of the retained placenta and/or improving subsequent reproductive performance; these treatments include manual removal, antibiotic and

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hormonal therapies [7]. Collagenase therapy using umbilical injections shows promise as a valuable approach in accelerating the expulsion of the retained placenta; however, it is not considered practical for widespread use because of its cost [8]. Few effective therapeutic approaches emphasize the importance of preventive treatment for retained placenta in cows. LeBlanc, et al. [9] have proposed that developing prevention strategies to fight against related diseases (e.g., subclinical mastitis, ketosis, rumen acidosis, and endometritis) would be of more value for a herd, and retained placenta is a prime example of a disease for which prevention strategies might well be a more efficient and possibly more effective long-term health management strategy than common treatment measures. A meta-analysis of studies investigating the effect of nutritional management during the dry period showed that vitamin E and selenium supplements might be effective in reducing the incidence of retained placenta [10]. Therefore, it is probable that appropriate prevention strategies represent effective options for retained placenta in dairy cows. Traditional Chinese medicine theory emphasizes the importance of disease prevention and early-stage treatment [11,12].

As a potential source of therapeutic aids, medicinal herbs play a pivotal role in health systems worldwide in both disease treatment and maintenance of proper health [13]. Many traditional herbs (administered alone or as a mixture) are often applied to prevent and treat retained placenta in dairy livestock [14–18]. *Sheng Hua Tang*, a classical herbal formula consisting of *Radix Angelicae sinensis*, *Ligustici rhizoma*, *Semen persicae*, *Zingiberis rhizoma*, and *Radix glycyrrhizae*, is usually used to activate blood circulation, increase uterine contraction, alleviate abdominal pain, and expel lochia in the treatment of postpartum women [19]. Current studies have demonstrated that *Sheng Hua Tang* has a beneficial effect on uterine involution by increasing contractile activity [20] and returning the uterus to its anteverted position [21]. *Sheng Hua Tang* has been used as a therapeutic aid in preventing and treating postpartum disease in female domestic animals. Lin, et al. [22] have shown that *Sheng Hua Tang* can play an important role in facilitating the return to normal reproductive function in postpartum sows. Li, et al. [23] have demonstrated that *Sheng Hua Tang* possesses certain beneficial effects and reduces the incidence of endometritis in cows.

To our knowledge, few data are available on the efficacy of *Sheng Hua Tang* in preventing retained placenta in dairy cattle. In the present study, a randomized controlled trial was designed to evaluate the effects of *Sheng Hua Tang* orally administered immediately after calving as a first step preventive treatment to reduce the incidence of retained placenta. This approach was recommended by an experienced traditional medicine practitioner. We tested whether *Sheng Hua Tang*, during the early postpartum stage, was an effective prevention strategy that would improve subsequent reproductive performance.

2. Materials and methods

2.1. *Sheng Hua Tang* preparation

The traditional formula for *Sheng Hua Tang* was used in our study. This formula is composed of five herbs: *Angelica*

sinensis, 120.0 g; *Ligusticum chuanxiong*, 45.0 g; *Prunus persica*, 30.0 g; *Zingiber officinale*, 10.0 g; and *Glycyrrhiza uralensis*, 10.0 g. These herbs were administered as a single dose to cows of 600-kg body weight (equal to 0.36 g crude drug per kg·body weight) [24]. The herb quality criteria were in accordance with the 2010 Chinese Pharmacopoeia [25]. In particular, all herbs were tested for heavy metals, microbial contamination, and residual pesticides, and the results met the safety standards of Chinese medicine in China. *Sheng Hua Tang* was prepared according to a standardized procedure to produce a decoction with a final concentration of 0.5 g crude herb per mL.

2.2. Animals

The trial was conducted over a period of 2 years (January 2010–January 2012) in a commercial dairy farm housing 520 Holstein dairy cows in northwestern China. All cows were housed in a free-stall barn with access to an adjoining sod-based paddock. A daily ration was offered as total mixed ration, which consisted of corn silage and hay as forage, with a corn gluten- and cottonseed meal-based concentrate and minerals and vitamins formulated to meet or exceed the nutritional requirements of early lactating dairy cows [26]. During the experimental period, all cows were fed ad libitum to allow approximately 5% feed refusals and had free access to high quality water. After calving, the cows were milked three times daily.

2.3. Enrollment criteria

The enrolled cows were clinically healthy, free of infectious or metabolic diseases, and 3 to 8 years of age (2–5 lactations) and had a current history of spontaneous delivery without assistance. They had a body condition score between 2.5 and 4.0 at the time of calving, according to the criteria specified in the protocol of Edmonson, et al. [27]. In addition, all cows involved in this trial were studied for only one lactation session.

2.4. Study design and treatment protocol

This study was designed as a randomized controlled study. The enrolled cows were randomly allocated to one of two groups based on their ear-tag numbers. The cows in the treatment group (odd ear-tag numbers) received an oral dose of *Sheng Hua Tang* of 430 mL per 600 kg·body weight (equal to 0.36 g crude herb per kg·body weight) within 2 to 4 hours after delivery once daily for three consecutive days. The cows in the control group (even ear-tag numbers) received no intervention. During the experimental period, an examination was performed every 2 hours to check for the expulsion of the placenta by visually examining the perineum. The uterine condition and placental expulsion were confirmed by vaginal and trans-rectal palpation at 12 hours postpartum. All cows failing to expel the retained placenta within 12 hours postpartum were treated with an oxytetracycline infusion administered to the uterus (2000 mg in 250 mL of normal saline), according to the usual management measures during the

experimental period. No attempts were made to manually remove the placenta within 12 hours postpartum.

All animals were examined at 3-week intervals with transrectal palpation until spontaneous estrus. Inseminations were performed year round. Artificial insemination (AI) was performed by an experienced inseminator with frozen–thawed semen 12 to 18 hours after observed standing estrus. Pregnancy was confirmed using transrectal palpation and ultrasonography between Days 40 and 48 after insemination. During the experimental period, all cows with diseases (endometritis, ovarian cysts, ovarian atrophy, and mastitis) were treated according to the usual herd management measures (e.g., intrauterine application of antibiotics, hormonal therapy). The date of parturition, date of estrus, dates of AIs, and puerperal diseases (e.g., ovarian cyst diseases, endometritis, mastitis, and displaced abomasum) were monitored and recorded by the same inseminator. To evaluate the reproductive performance of the cows treated with *Sheng Hua Tang*, the calving-to-first-service interval, calving-to-conception interval (days not pregnant to pregnancy), first AI conception proportion, service per conception, and proportion of cows pregnant within 180 days postpartum were measured. The animals were reinseminated a maximum of six times (until Day-180 postpartum). Then, if they were not pregnant, they were excluded from the study and culled because of infertility.

2.5. Efficacy measurements

The primary measure of outcome was based on the effect of *Sheng Hua Tang* on the incidence of retained placenta. A retained placenta was diagnosed if it was not completely expelled within 12 hours after delivery. The secondary outcome measures included the first AI conception proportion, the calving-to-first-service interval, calving-to-conception interval (days not pregnant to pregnancy), service per conception, and proportion of cows that were pregnant within 180 days postpartum.

2.6. Exclusion criteria

During the experimental period, the cows that had diseases such as metabolic disorders, lameness, and displaced abomasum were withdrawn from the trial. Enrolled cows were excluded if the cows were moved out of the dairy farm at which we conducted the clinical trial (e.g., sold by the farm).

2.7. Statistical analysis

The data were analyzed using SPSS for Windows (version 21.0; IBM SPSS Statistics, NY, USA). A one-way ANOVA and a Kruskal–Wallis rank sum test, as appropriate, were used to compare baseline characteristics between the two groups. The proportion of cows that developed retained placenta within 12 hours after delivery was tested with a chi-squared test.

The interval from calving to first service and conception were compared between the cows in the *Sheng Hua Tang* group and the cows in the control group with a Cox

proportional hazards regression. Values were censored, if observations were terminated for reasons beyond the investigator's control (e.g., data for cattle that were never inseminated and conceived). Puerperal diseases (e.g., ovarian cysts, mastitis, and endometritis) were pooled into a single category, as the number of occurrences was small. For continuous variables, a semi-parametric model, the Cox proportional hazard regression, was applied. To test the influence of selected factors on the treatment (group), parity, puerperal diseases, and service per conception were included in the model. The survival analysis involved a regression that included the first AI conception proportion and the proportion of cows that were pregnant within 180 days postpartum. Observations were censored, if a cow died or was culled or at the time the study was discontinued (i.e., 180 days after calving). The experimental data were presented as the mean $\pm$ standard deviation, and P-value of 0.05 or less was considered significant for all tests.

3. Results

3.1. Participant characteristics

Before the clinical trial, the annual incidence of retained placenta at the dairy farm in which the clinical trial was conducted was 21.0%, 30.8%, and 20.3% in 2007, 2008, and 2009, respectively. During the experimental period, 357 cows were included in this study according to the enrollment criteria. Of these 357 cows, 175 cows were randomly assigned to the *Sheng Hua Tang* group and 182 to the control group. The baseline demographic characteristics are shown in Table 1. There were no differences in any individual characteristics, including the body condition score at calving, mean parity, mean age, and the proportion of twins.

During the subsequent observation process, one cow with serious lameness, one cow that died accidentally because of a collision with a fence, and four cows that were sold to another dairy farm were excluded from the *Sheng Hua Tang* group. Two cows with serious lameness, three with a displaced abomasum, and two that were sold to another dairy farm were excluded from the control group. Finally, 169 cows in the *Sheng Hua Tang* group and 175 cows in the control group were eligible for reproductive performance analysis.

3.2. Effect of *Sheng Hua Tang* on the incidence of retained placenta

The primary outcome measure depended on the effect of *Sheng Hua Tang* on the incidence of retained placenta

Table 1
Characteristics of the enrolled animals.

Characteristic	Treatment group	Control group
BCS	3.41 $\pm$ 0.63	3.52 $\pm$ 0.45
Mean parity	3.37 $\pm$ 1.03	3.52 $\pm$ 1.15
Mean age	5.68 $\pm$ 1.21	5.73 $\pm$ 1.34
Twin proportion (n/N)	5 (175)	6 (182)

Abbreviation: BCS, body condition score.

Table 2Effect of *Sheng Hua Tang* on the reproductive parameters of dairy cows.

Reproductive parameter	<i>Sheng Hua Tang</i> group (n = 169)	Control group (n = 175)	Difference (P-value)
Calving-to-first-service interval ^a , days	73.2 ± 25.1	81.9 ± 32.8	<0.01
First AI conception proportion ^b , %	60.4	41.1	0.01
Service per conception ^c , days	1.5 ± 0.8	1.9 ± 1.0	<0.01
Calving-to-conception interval ^d , days	93.4 ± 38.8	114.6 ± 42.9	<0.01
Proportion of cows that were pregnant within 180 days postpartum ^e , %	88.2	80.6	0.07

Abbreviation: AI, artificial insemination.

^a Calving-to-first-service interval was defined as the number of days from calving to the first service.^b Service per conception was defined as the average number of artificial inseminations until pregnancy.^c First AI conception proportion was defined as the number of cows that were pregnant in first service divided by the total number of inseminations.^d The calving-to-conception intervals were analyzed based on the cows that were pregnant within 180 days postpartum.^e Proportion of cows that were pregnant within 180 days expressed as a proportion of the inseminated animals in the respective group.

after delivery. According to the above-stated criterion, 168 cows in the *Sheng Hua Tang* group released the placenta spontaneously within 12 hours after calving, whereas seven cows (4.0%) were diagnosed with retained placenta. In the control group, 151 cows expelled the placenta spontaneously, and retained placenta occurred in 31 of 182 cows (17.0%). The difference between the two groups was statistically significant ($P < 0.01$).

3.3. Effect of *Sheng Hua Tang* on subsequent reproductive performance

The long-term efficacy of *Sheng Hua Tang* was evident from the facilitation of the return to normal production function; these results are shown in Table 2. The interval from calving to first service was shorter in the *Sheng Hua Tang* group than control group ($P < 0.01$). The number of services per conception was significantly less in the *Sheng Hua Tang* group than in the control group ($P < 0.01$). The first AI conception proportion of cows in the *Sheng Hua Tang* group increased significantly compared with the cows in the control group ($P = 0.01$). The calving-to-conception interval of the cows was significantly shorter in the *Sheng Hua Tang* group than the control group ($P < 0.01$). The survival curves differed between the *Sheng Hua Tang* group and the control group in terms of the calving-to-conception interval ($P < 0.001$; Fig. 1). The difference between the two groups in the proportion of cows that were pregnant within 180 days postpartum approached statistical significance ($P = 0.07$).

4. Discussion

Traditional postpartum practices are believed to be beneficial in reducing the risk of postpartum diseases and facilitating a return to normal reproductive function [18,28–30]. *Sheng Hua Tang* has been used a therapeutic aid in preventing and treating postpartum disease in clinical practice for centuries [19,29]. In the present study, we investigated whether the oral administration of *Sheng Hua Tang* within 2 to 4 hours after delivery was effective as a preventive treatment for reducing the risk of retained placenta. The results suggest that *Sheng Hua Tang* is effective at reducing the incidence of retained placenta and improving the reproductive performance of cows during the early postpartum stage.

In the theory of traditional Chinese veterinary medicine, retained placenta belongs to the category of blood stasis syndrome. Blood stasis is an important underlying pathology of many disease processes that manifest as hematological disorders, such as hemorrhage, congestion, thrombosis, local ischemia, and tissue changes [31], and it has been considered a key aspect of pathogenesis in postpartum diseases [19,32]. Many herbal agents with blood circulation-activating activities are well documented to reduce the risk of postpartum diseases and improve the health-related quality of life in the postpartum period [28–30]. *Sheng Hua Tang* is a good example of a herbal formula that activates the blood circulation and removes blood stasis, thereby ameliorating postpartum diseases [19,29]. Our study showed that *Sheng Hua Tang* as a prophylactic strategy could effectively reduce the incidence of retained placenta in dairy cows. Previous studies have demonstrated that *Sheng Hua Tang* exerts a wide range of biological effects, modulating and restoring regular uterine contractions [20], facilitating uterine involution [21], inducing anti-

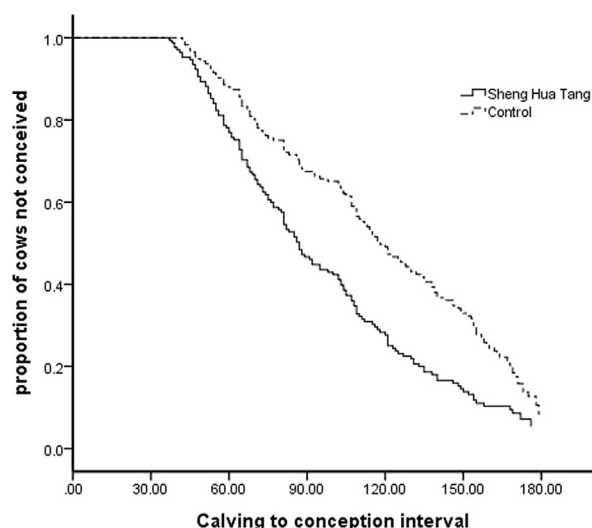


Fig. 1. The proportion of cows that did not conceive within 180 days postpartum (survival analyses). Survival curves are presented for cows in the *Sheng Hua Tang* (received an oral dose of 430 mL per 600 kg body weight [equal to 0.36 g crude herb per kg body weight] after delivery once daily for three consecutive days) and control (received no intervention) groups.

inflammatory and analgesia responses [33], modulating the immune response [34,35], and inducing hematological responses [36]. These effects might act synergistically to help lower the risk of puerperal disease. It was speculated that *Sheng Hua Tang* might possess beneficial effects to improve overall puerperal health and facilitate a return to normal physical status to help lower the risk of retained placenta. Before the clinical trial, the annual incidence of retained placenta was greater than 20.0% at the dairy farm at which the clinical trial was conducted. In the present study, the incidence of retained placenta was 4.0% in treatment group and *Sheng Hua Tang* resulted in a significant difference in the placental retention proportion as compared with controls. Therefore, the use of *Sheng Hua Tang* immediately after delivery should be considered an effective prophylactic strategy for retained placenta in dairy cows.

The reproductive performance of a dairy herd has a significant effect on the profitability of that herd. Reproductive efficiency and profitability are maximized by the timely achievement of a subsequent pregnancy [37]. Therefore, effective postpartum health care should aim to prepare the cows for subsequent pregnancy at a biologically optimal time by improving overall puerperal health during the early postpartum stage [38]. In the present study, the early administration of *Sheng Hua Tang* postpartum advanced the calving-to-first-service interval and shortened the calving-to-conception interval. It has been demonstrated that the number of days to first service has a significant impact on the first AI conception proportion in dairy cows [39]. *Sheng Hua Tang* effectively increased the proportion of first AI conception. The pregnancy rate is a good overall measure of reproductive performance. This result suggests that *Sheng Hua Tang* effectively improved the reproductive performance of the cows. Previous studies have demonstrated that *Sheng Hua Tang* can facilitate uterine involution in postpartum women [21]. Kindahl, et al. [40] have proposed that uterine involution is inherently associated with improved ovarian function; they must both be achieved for a new subsequent pregnancy to occur. Therefore, *Sheng Hua Tang* might improve the reproductive performance of the cows by facilitating uterine involution in the postpartum stage.

The reproductive performance of a dairy herd is associated with an increased conception rate at a desired calving interval [33]. In general, calving-to-conception interval decreases when the pregnancy rate increases. In the present study, *Sheng Hua Tang* effectively reduced the number of services per conception, although the proportion of cows that were pregnant at 180 days postpartum did not reach a significant difference in the two groups. Thus, this therapeutic option would effectively increase potential savings because of result in fewer inseminations, shortening in calving-to-conception interval and reduction in the placental retention proportion. In addition, the use of *Sheng Hua Tang* to reduce the incidence of retained placenta and improve fertility in cows is interesting and warranted in a time where restrictions on the use of antimicrobial drugs are increasingly imposed on animal agriculture for lack of residues [41,42]. Therefore, the *Sheng Hua Tang* treatment used in the present study should be considered an effective prophylactic strategy for retained placenta in

dairy cows. The findings of our study also provide a good starting point for further research. However, because of the small number of animals involved in this study, these findings are considered preliminary, and a large, multiple-site field trial should be conducted in the future to test our hypothesis. Future investigations will focus on the broader involvement of *Sheng Hua Tang* in the mechanisms underlying its therapeutic effects.

4.1. Conclusions

The use of *Sheng Hua Tang* as a preventive treatment after delivery produced beneficial effects in reducing the incidence of retained placenta and improving reproductive performance in dairy cows. This treatment measure could represent a practical preventive strategy because of the potential cost savings by lowering the risk of retained placenta, decreasing the calving-to-first-service interval and the calving-to-conception interval, reducing service per conception and elevating the first AI conception proportion and the proportion of cows that were pregnant within 180 days postpartum. *Sheng Hua Tang* potentially provides a preventive strategy for bovine postpartum care.

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